WRS Evolution

Requirements Elicitation

CptS 484: Software Requirements
10/18/2026

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Revision History

Date	Version	Changes	Editor

10/17/2026	1.0	Initial Draft	Jackson Pickett, Faisal Kherwish, and Hank Corrion
10/17/2026	1.1	Updated Section 2 and 3	Faisal Kherwish
10/18/2026	1.2	Updated Section 1	Jackson Pickett
10/18/2026	1.3	Updated Section 4	Hank Corrion
10/18/2026	1.4	Final Proofing	Jackson Pickett, Faisal Kherwish, and Hank Corrion

[1] Introduction

1.1 Purpose

The purpose of project THEIA is to help visually impaired people navigate environments safely and independently. Navigating unfamiliar buildings can be difficult for people with visual impairments, especially when trying to find different rooms, avoid obstacles, and move between

floors. THEIA aims to address these challenges by providing an app that gives step by step instructions, location tracking, obstacle warnings, and emergency assistance.

1.2 Scope

Project THEIA focuses on developing an app that helps visually impaired people navigate indoor environments such as buildings, classrooms, offices, hallways, and other facilities. The app is intended to help users determine their current location, select destinations, receive step by step instructions, and avoid obstacles.

THEIA will primarily focus on indoor navigation, while outdoor navigation between buildings will be handled through existing tools. The current phase of the project focuses on requirements engineering, documentation, and creating a prototype.

1.3 Objectives and Success Criteria

The main objective of Project THEIA is to improve the independence, safety, and accessibility of indoor navigation for visually impaired people.

The primary objectives of THEIA are:

- Allowing users to navigate indoor environments
- Providing accurate location information and step by step directions
- Helping users reach their destinations safely
- Warning users of potential obstacles
- Providing assistance when users become lost or are in an emergency
- Supporting accessibility features

1.4 Definitions, Acronyms, and Abbreviations

THEIA: An app designed to assist visually impaired people with indoor navigation
WRS (World, Requirements, and Specification): A requirements engineering model used to describe the problem domain, system requirements, and specifications needed to address identified problems.
Functional Requirement (FR): A requirement that describes a function or behavior the system must provide.
Non-Functional Requirement (NFR): A requirement that describes the quality, performance, accessibility, safety, or other constraints of a system.
Preliminary Definition (PD): The original description of the THEIA project that provides the initial domain information, objectives, and system requirements.
AS-IS Scenario: A description of how users currently perform an activity or experience a problem without the proposed system.
TO-BE Scenario: A description of how users would perform an activity or solve a problem with the proposed system.

Stakeholder: A person or organization that is affected by or has an interest in the development and operation of THEIA.
Traceability: The ability to connect identified problems and goals to the requirements intended to address them.
Prototype: An early representation of the app used to demonstrate its proposed interface and functionality.

1.5 Overview

Section 1 introduces the project and describes the project's purpose, scope, objectives, and terminology.

Section 2 identifies issues found in the preliminary definition of THEIA, including problems with the domain description, functional requirements, and non-functional requirements. Each issue includes possible solutions, the team's selected decision, and the reasoning behind the decision.

Section 3 presents the improved WRS model. This section identifies the problems the app is intended to solve, establishes project goals, defines stakeholders, and presents the improved functional and non-functional requirements.

Section 4 covers the preliminary prototype and user manual, which demonstrates how users are expected to interact with THEIA.

Section 5 describes the traceability between identified problems, project goals, objectives, and system requirements to ensure that the requirements support the overall purpose of the app.

[2] Issues with the Preliminary Definition Given

2.1. Domain Issues

Domain Issue ID	**Domain Issue Description**
DI1	**PD_ID PD1, PD4**: Indoor navigation within the same or connected buildings. Ambiguous. The Project Specification describes reaching classes across campus, which often requires going outdoors.

Option 1	Indoor navigation with hand-off to an existing outdoor navigation app between buildings
Option 2	Indoor only, connected buildings only.
Choice	Option 1
Rationale	Outdoor GPS navigation is already solved by existing apps; a hand-off supports campus trips while keeping scope on indoors.
Satisfied by	ID1, ID2, FR6

Domain Issue ID	**Domain Issue Description**
DI2	**PD_ID PD1**: Floors with classrooms, offices, bathrooms, lounges, elevators. Incomplete. Stairs, doors, exits and hazards are missing, and the map source is never stated.
Option 1	Assume detailed maps already exist.
Option 2	Maps maintained by building administrators.
Choice	Option 2
Rationale	Facilities staff own accurate floor plans and know about closures.
Satisfied by	ID2, ID3, ID4

Domain Issue ID	**Domain Issue Description**
DI3	**PD_ID PD2, PD3**: A blind person; accessibility staff and police. Ambiguous / Incomplete. Ignores low-vision users; secondary roles are undefined and building administrators

	are missing.
Option 1	Keep the stakeholders as given.
Option 2	Include all visually impaired users and define every stakeholder's role.
Choice	Option 2
Rationale	Matches the Project Specification; undefined roles lead to missing requirements.
Satisfied by	Stakeholders, FR8, FR11

Domain Issue ID	**Domain Issue Description**
DI4	**PD_ID PD3**: A caretaker sets the app's configuration. Conflict. This undermines the user's independence and privacy.
Option 1	User controls the app and invites a caretaker with limited permissions.
Option 2	Caretaker controls configuration.
Choice	Option 1
Rationale	Keeps the user independent; location is shared only in emergencies.
Satisfied by	FR11, NFR11

2.2. Functional Requirements Issues

FR Issue ID	**Description**
FRI1	**PFR_ID PFR1**: Accept the destination from the user. Ambiguous. Input method is unstated, and names like "Room 201" may exist in several

	buildings.
Option 1	Exact room numbers only.
Option 2	Voice or typed input, with a clarifying question when several places match.
Choice	Option 2
Rationale	Users think in names and categories, not room numbers.
Satisfied by	FR1

FR Issue ID	**Description**
FRI2	**PFR_ID PFR2**: Figure out routes to the destination. Incomplete. No requirement finds the user's location, and GPS fails indoors.
Option 1	Hybrid of beacons, Wi-Fi, sign scanning and step tracking.
Option 2	Bluetooth beacons only.
Choice	Option 1
Rationale	No single method works in every building.
Satisfied by	FR2, FR9, NFR8

FR Issue ID	**Description**
FRI3	**PFR_ID PFR3**: Walk 2 minutes or 30 steps. Inconsistent. Mixes time and steps; both are inaccurate without calibration.
Option 1	Time only.
Option 2	Calibrated steps or distance plus

	landmark cues.
Choice	Option 2
Rationale	Accurate counts, confirmed by landmarks.
Satisfied by	FR4

FR Issue ID	**Description**
FRI4	**PFR_ID PFR4**: Stop at the right place to turn. Incomplete. No advance warning, missed-turn recovery, or floor-change guidance.
Option 1	Advance warning, vibration, rerouting, and elevator/stair guidance.
Option 2	A single announcement at the turn.
Choice	Option 1
Rationale	Missed turns and floor changes are where users most need help.
Satisfied by	FR5, FR6

FR Issue ID	**Description**
FRI5	**PFR_ID PFR5**: Detect obstacles. Ambiguous. Obstacle types and method are unstated; implies replacing the cane.
Option 1	Camera detection only.
Option 2	Camera plus known map hazards, as a supplement to the cane or dog.
Choice	Option 2

Rationale	Works whether the phone is held or pocketed; sets honest limits.
Satisfied by	FR7, NFR1, ID5

FR Issue ID	**Description**
FRI6	**PFR_ID PFR6**: Emergency calls after a fall or lost location. Unsound. Being lost is not an emergency, and false fall alarms cannot be cancelled.
Option 1	Call emergency services for every trigger.
Option 2	Tiered response with a cancel window.
Choice	Option 2
Rationale	Fast help when hurt, without false emergency calls.
Satisfied by	FR8

FR Issue ID	**Description**
FRI7	**PFR_ID PFR1, PFR7**: Suggestions from the user's schedule. Redundant. Both describe the same feature.
Option 1	Keep two separate features.
Option 2	Merge into one feature with an off switch.
Choice	Option 2
Rationale	Simpler and less distracting while walking.
Satisfied by	FR10

2.3. Non-Functional Requirements (NFR) Issues

NFR Issue ID	**Description**
NFRI1	**PNFR_ID PNFR1**: Help the user safely navigate. Ambiguous. Safety is not measurable as written.
Option 1	Keep it as a general goal.
Option 2	Measurable criteria: warning time and a stop-don't-guess rule.
Choice	Option 2
Rationale	A requirement that cannot be tested cannot be verified.
Satisfied by	NFR1

NFR Issue ID	**Description**
NFRI2	**PNFR_ID PNFR2, PNFR3**: Fastest route; most comfortable route. Inconsistent. Both cannot hold, and "comfortable" is undefined.
Option 1	User-selected mode plus a comfort profile (turns, stairs, crowds).
Option 2	Always the fastest route.
Choice	Option 1
Rationale	Resolves the conflict per user and makes comfort measurable.
Satisfied by	FR3, NFR2, NFR3

NFR Issue ID	**Description**

NFRI3	**PNFR_ID PNFR4**: Usable for blind people. Ambiguous. No usability targets; voice alone fails in noisy or quiet places.
Option 1	Informal team judgment, voice only.
Option 2	Multiple input methods and measured usability with visually impaired testers.
Choice	Option 2
Rationale	Only real users can judge non-visual usability.
Satisfied by	NFR4

NFR Issue ID	**Description**
NFRI4	**PNFR_ID PNFR5**: The system shall be ubiquitous. Ambiguous. Could mean any building, any phone, or offline use.
Option 1	Works in every building.
Option 2	Mapped buildings, common phones, offline, hands-free.
Choice	Option 2
Rationale	Realistic and testable.
Satisfied by	NFR5

NFR Issue ID	**Description**
NFRI5	**PNFR_ID PNFR6, PNFR7**: Customizable to every user; easily extensible. Ambiguous. Neither is bounded or measurable.
Option 1	A defined list of voice-adjustable

	settings and a modular design.
Option 2	Keep the general statements.
Choice	Option 1
Rationale	Both become buildable and testable.
Satisfied by	FR11, NFR6

NFR Issue ID	**Description**
NFRI6	**No PNFR_ID**. Accuracy, response time, battery and privacy are not addressed.
Option 1	Defer to later phases.
Option 2	Add them now with measurable targets.
Choice	Option 2
Rationale	Each is a safety or trust risk for the user.
Satisfied by	NFR7, NFR8

[3] WRS

3.1. W

3.1.1. Problem

Problem ID	**Problem Description**
P1	Blind users can't tell where they are indoors.
P2	Users don't know which way to go or when to turn.

P3	Obstacles and hazards cause collisions and falls.
P4	Unfamiliar routes make users late.
P5	Changing floors or buildings is hard without sight.
P6	Lost or injured users can't easily get help.
P7	Users have different needs and preferences.
P8	Standard apps aren't usable without sight.
P9	Location and schedule data are sensitive.

3.1.2 Goal

Goal ID	**Goal Description**	**Backward Traceability**	**Forward Traceability**
G1	Navigate independently indoors.	P1, P2, P5	IFRO1–IFRO4, IFRO7
G2	Arrive on time.	P2, P4, P5	IFRO3, IFRO4, IFRO8, INFRO2
G3	Navigate safely.	P3, P6	IFRO5, INFRO1
G4	Get help quickly in emergencies.	P6	IFRO6, INFRO1
G5	Be usable and adaptable for every VI user.	P7, P8	INFRO3–INFRO6
G6	Be accurate, reliable and private.	P1, P9	INFRO7

3.1.3. Improved understanding of Domain, Stakeholders, Functional, and Non-Functional Objectives

3.1.3.1 Improved Domain

Improved Domain ID	**Improved Domain Description**
ID1	Mapped buildings: floors, hallways, rooms, restrooms, lounges. (DI1)
ID2	Connectors: elevators, stairs, doors, exits; outdoor segments handed off to an outdoor app. (DI1, DI2)
ID3	Hazards: obstacles, moving people, overhead objects, drop-offs, closures. (DI2)
ID4	Building administrators maintain the map data. (DI2)
ID5	The user has a smartphone and still uses a cane or guide dog. (FRI5)

3.1.3.2. Stakeholders

• Primary users: blind and visually impaired people. (DI3)
• Caretakers: invited by the user; emergency contacts. (DI4)
• Building administrators: maintain maps and report closures. (DI2)
• Accessibility staff and O&M instructors: recommend the app and train users. (DI3)
• Campus security / emergency services: respond to escalated emergencies. (DI3)

3.1.3.3. Improved Functional Objectives

Improved FR Objective ID	**Objective Description**
IFRO1	Accept the destination.
IFRO2	Determine the user's location.
IFRO3	Plan and offer routes.

IFRO4	Give step-by-step guidance, including floor changes.
IFRO5	Warn of obstacles and hazards.
IFRO6	Provide emergency assistance.
IFRO7	Let the user explore nearby places.
IFRO8	Suggest the next destination from the schedule.

3.1.3.4. Improved Non-Functional Objectives

Improved NFR Objective ID	**Objective Description**	**Alleviates Problems**	**Achieves Goals**
INFRO1	Safety is never traded away.	P3, P6	G3, G4
INFRO2	Fastest route fits the user's walking speed.	P4	G2
INFRO3	Routes match the user's comfort profile.	P7	G5
INFRO4	Fully usable without sight.	P8	G5
INFRO5	Works on common phones, offline, hands-free.	P8	G5
INFRO6	Customizable and extensible.	P7	G5
INFRO7	Accurate, responsive, battery-efficient and private.	P1, P9	G6

3.2. RS

3.2.1. Functional Requirements

FR ID	**Description**
FR1	Accept the destination by voice or text; ask to clarify if ambiguous.
Satisfies Functional Requirement Issue	FRI1
Satisfies Objectives	IFRO1
Satisfied by prototype feature	Destination Search

FR ID	**Description**
FR2	Find the user's location and announce it on "Where am I?".
Satisfies Functional Requirement Issue	FRI2
Satisfies Objectives	IFRO2
Satisfied by prototype feature	Where Am I

FR ID	**Description**
FR3	Offer up to three routes with a recommended default.
Satisfies Functional Requirement Issue	NFRI2
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Route Options

FR ID	Description
FR4	Give distances in calibrated steps or feet, with landmarks.
Satisfies Functional Requirement Issue	FRI3
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Guided Navigation

FR ID	Description
FR5	Warn before turns by voice and vibration; reroute if off route.
Satisfies Functional Requirement Issue	FRI4
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Guided Navigation

FR ID	Description
FR6	Guide through elevators and stairs; hand off outdoor segments.
Satisfies Functional Requirement Issue	FRI4, DI1
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Guided Navigation

FR ID	Description
FR7	Warn of obstacles with direction, distance and action.
Satisfies Functional Requirement	FRI5

Issue	
Satisfies Objectives	IFRO5
Satisfied by prototype feature	Obstacle Alerts

FR ID	**Description**
FR8	After a fall, check on the user, then alert emergency contacts; manual SOS.
Satisfies Functional Requirement Issue	FRI6
Satisfies Objectives	IFRO6
Satisfied by prototype feature	Emergency / SOS

FR ID	**Description**
FR9	Describe nearby places and read signs aloud.
Satisfies Functional Requirement Issue	FRI2
Satisfies Objectives	IFRO7
Satisfied by prototype feature	Explore

FR ID	**Description**
FR10	Suggest the next destination from the calendar; can be turned off.
Satisfies Functional Requirement Issue	FRI7
Satisfies Objectives	IFRO8
Satisfied by prototype feature	Suggestions

FR ID	Description
FR11	Change settings by voice; invite a caretaker.
Satisfies Functional Requirement Issue	DI4, NFRI5
Satisfies Objectives	INFRO6
Satisfied by prototype feature	Settings

3.2.2. Non-functional Requirements

NFR ID	Nonfunctional Requirement 1
NFR1	Obstacle warnings at least 2 s before contact; stop the user when position is uncertain.
Operationalized Functional Requirements	FR2, FR5, FR7
Satisfies Nonfunctional Requirement Issue	NFRI1
Satisfies Non-functional Objective	INFRO1
Constrains	P3, P6
Satisfied by prototype feature	Guided Navigation, Obstacle Alerts

NFR ID	Nonfunctional Requirement 1
NFR2	Travel time estimates within 20% of actual.
Operationalized Functional Requirements	FR3
Satisfies Nonfunctional Requirement Issue	NFRI2

Satisfies Non-functional Objective	INFRO2
Constrains	P4
Satisfied by prototype feature	Route Options

NFR ID	**Nonfunctional Requirement 1**
NFR3	Routes scored by the user's comfort profile.
Operationalized Functional Requirements	FR3
Satisfies Nonfunctional Requirement Issue	NFRI2
Satisfies Non-functional Objective	INFRO3
Constrains	P7
Satisfied by prototype feature	Route Options

NFR ID	**Nonfunctional Requirement 1**
NFR4	Works with VoiceOver/TalkBack; destination set in 3 or fewer steps.
Operationalized Functional Requirements	FR1, FR5, FR11
Satisfies Nonfunctional Requirement Issue	NFRI3
Satisfies Non-functional Objective	INFRO4
Constrains	P8
Satisfied by prototype feature	All features

NFR ID	**Nonfunctional Requirement 1**

NFR5	Runs on current iOS/Android, offline, with the phone in a pocket.
Operationalized Functional Requirements	All FRs
Satisfies Nonfunctional Requirement Issue	NFRI4
Satisfies Non-functional Objective	INFRO5
Constrains	P8
Satisfied by prototype feature	All features

NFR ID	**Nonfunctional Requirement 1**
NFR6	All settings adjustable by voice; new languages and sensors added without changing core code.
Operationalized Functional Requirements	FR2, FR7, FR11
Satisfies Nonfunctional Requirement Issue	NFRI5
Satisfies Non-functional Objective	INFRO6
Constrains	P7
Satisfied by prototype feature	Settings

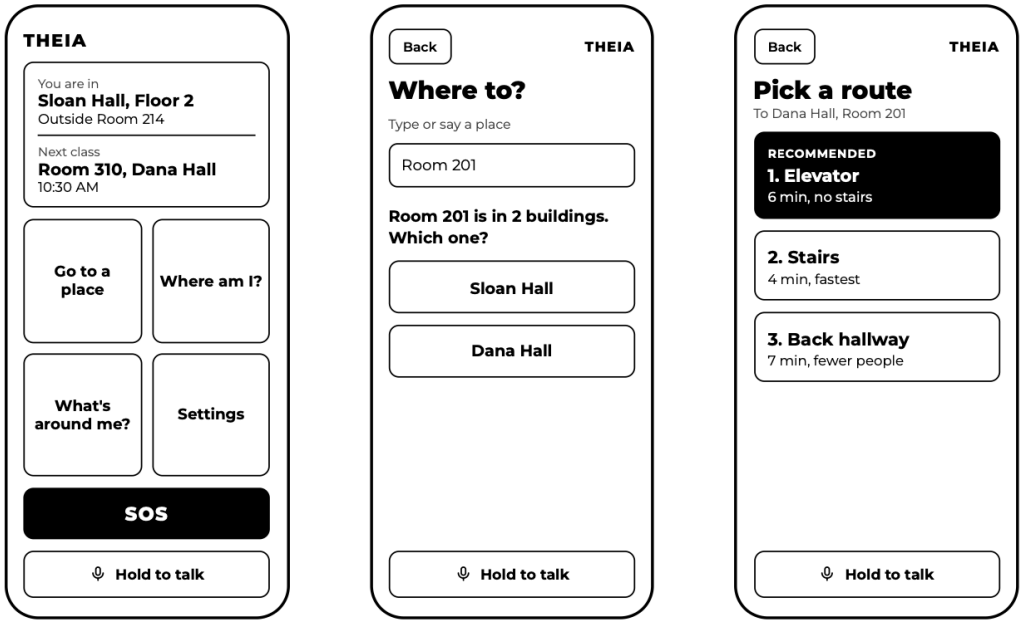
NFR ID	**Nonfunctional Requirement 1**
NFR7	Position within 2 m (90% of the time); instructions within 1 s; at most 15% battery per hour.
Operationalized Functional Requirements	FR2, FR5, FR6, FR7
Satisfies Nonfunctional Requirement Issue	NFRI6

Satisfies Non-functional Objective	INFRO7
Constrains	P1, P2, P3
Satisfied by prototype feature	Where Am I, Guided Navigation

NFR ID	**Nonfunctional Requirement 1**
NFR8	Location and calendar data stay on the device unless shared by consent or in an emergency.
Operationalized Functional Requirements	FR2, FR8, FR10
Satisfies Nonfunctional Requirement Issue	NFRI6, DI4
Satisfies Non-functional Objective	INFRO7
Constrains	P9
Satisfied by prototype feature	Emergency / SOS, Suggestions

[4] Preliminary Prototype and User Manual

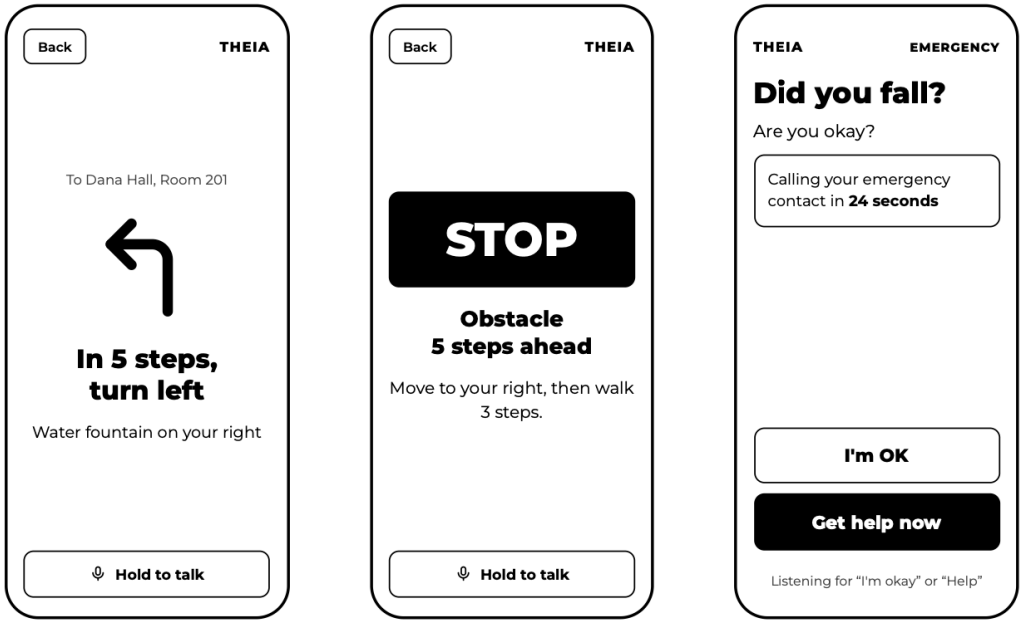
The preliminary prototype of THEIA is a set of sketched screens with the voice commands for each feature, shown in Figures 4.1 and 4.2. Each screen is labeled with the functional requirement it covers. No code was written for Phase I. Section 4.1 is the user manual, which explains how to use each feature.



1. Home / Where Am I (FR2)
User: "Where am I?" THEIA: "You're in Sloan Hall, floor 2, outside room 214."

2. Destination Search (FR1)
User: "Take me to room 201" THEIA: "Room 201 is in Sloan Hall and Dana Hall. Which one?"

3. Route Options (FR3)
User: "Option one" THEIA: "Route one is recommended. Elevator, 6 minutes, no stairs."



4. Guided Navigation (FR4, FR5, FR6)
THEIA: "In 5 steps, turn left." The phone vibrates before each turn.

5. Obstacle Alerts (FR7)
THEIA: "Stop. Obstacle 5 steps ahead. Move to your right and walk 3 steps."

6. Emergency / SOS (FR8)
User: "I'm okay" or "Help." No answer in 30 seconds calls the emergency contact.

Figure 4.1: THEIA prototype screens 1 to 6

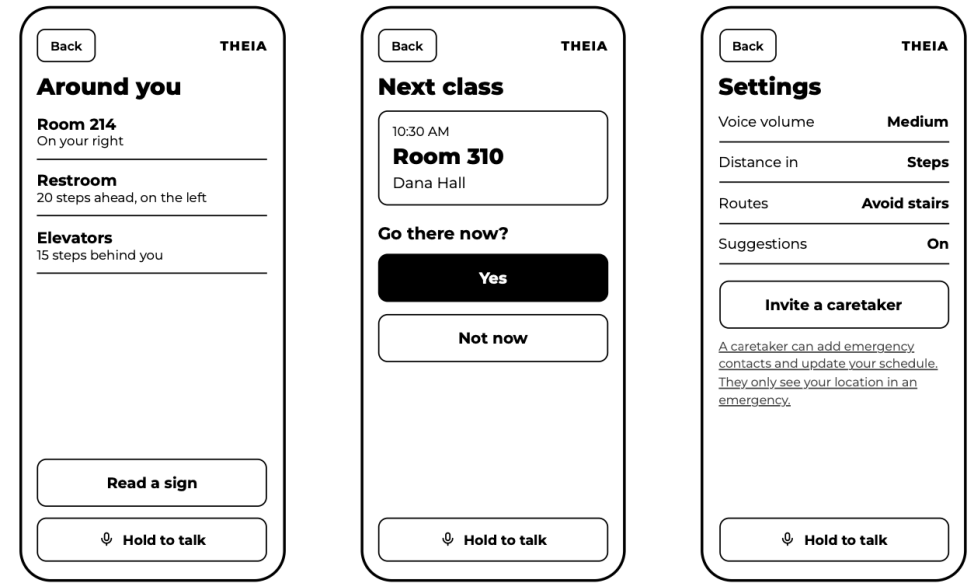


Figure 4.2: THEIA prototype screens 7 to 9

4.1 User Manual

This user manual explains how to use each feature of THEIA. The steps follow the scenarios from our presentation, and the screen numbers refer to Figures 4.1 and 4.2.

4.1.1 Getting Started

1. Open THEIA. The home screen shows where you are and your next class (Screen 1).
2. To give a command, hold the Hold to talk button, say the command, then let go. You can also tap the buttons on the screen.
3. THEIA works with VoiceOver and TalkBack, and it reads every instruction out loud.
4. Keep using your cane or guide dog. THEIA is meant to help alongside them, not replace them.

4.1.2 Getting to a Room (Screens 2, 3 and 4)

1. Hold Hold to talk and say where you want to go, for example “Take me to room 201.” You can also type it.
2. If the place exists in more than one building, THEIA asks which one. Say the building name.
3. THEIA reads up to three routes. The recommended route comes first. Say the number of the route you want, for example “Option one.”
4. Follow THEIA’s directions. It tells you how many steps to walk and what landmarks you will pass.

5. Before each turn, your phone vibrates and THEIA warns you. If you miss a turn, THEIA tells you how to get back on route.
6. THEIA tells you where the elevator or stairs are and which floor to go to. If part of the trip is outside, THEIA opens your outdoor map app.

4.1.3 Finding Out Where You Are (Screen 1)

1. Say “Where am I?” or press Where am I? on the home screen.
2. THEIA tells you the building, floor and nearest room.
3. If THEIA is not sure where you are, it tells you to stop. Stay where you are until THEIA finds your location again.

4.1.4 When THEIA Warns You About an Obstacle (Screen 5)

1. When THEIA says “Stop,” stop walking.
2. Listen for how far away the obstacle is and how to get around it, for example “Move to your right and walk 3 steps.”
3. Once you are past it, THEIA goes back to your directions.

4.1.5 If You Fall or Need Help (Screen 6)

1. If THEIA detects a fall, it asks “Are you okay?”
2. To cancel, say “I’m okay” or tap I’m OK. Nothing is sent.
3. To get help right away, say “Help” or tap Get help now.
4. If you do not answer within 30 seconds, THEIA calls your emergency contact and sends your location.
5. To ask for help at any time, say “SOS” or press SOS on the home screen.

4.1.6 Exploring Nearby Places (Screen 7)

1. Say “What’s around me?” or press What’s around me? on the home screen.
2. THEIA lists nearby rooms and places and tells you which direction each one is.
3. To read a sign, say “Read this sign” or press Read a sign, then point your phone’s camera at the sign.

4.1.7 Class Suggestions (Screen 8)

1. Before each class, THEIA checks your schedule and asks if you want to go there.
2. Say “Yes” to start directions, or “Not now” to skip.
3. To turn suggestions off, say “Turn off suggestions” or change it in Settings.

4.1.8 Changing Settings and Inviting a Caretaker (Screen 9)

1. Open Settings from the home screen, or just say what you want to change, for example “Make it louder” or “Avoid stairs.”

2. You can change the voice volume, whether distances are given in steps or feet, your route preferences, and class suggestions.
3. To add a caretaker, say “Invite a caretaker” and choose a contact.
4. A caretaker can add emergency contacts and update your class schedule. They only see your location in an emergency.

4.1.9 Voice Commands

The table below lists the voice commands THEIA understands.

Voice Command	**What It Does**
“Take me to [place]”	Starts directions to a place
“Where am I?”	Tells you your building, floor and nearest room
“Option one,” “Option two,” “Option three”	Picks a route
“What’s around me?”	Lists nearby places
“Read this sign”	Reads a sign out loud
“I’m okay”	Cancels a fall alert
“Help”	Calls your emergency contact
“SOS”	Asks for help at any time
“Yes” or “Not now”	Answers a class suggestion
“Turn off suggestions”	Stops class suggestions
“Make it louder”	Raises the voice volume
“Use feet”	Gives distances in feet instead of steps
“Avoid stairs”	Chooses routes without stairs
“Invite a caretaker”	Sends a caretaker invite

[5] Traceability

Traceability is used to show how the problems identified in Project THEIA connect to the project's goals and system requirements. It helps ensure that each requirement serves a purpose and that the original problems are addressed by the proposed system. Forward traceability follows the relationships from problems and goals to the requirements that address them, while backward traceability connects requirements to the objectives and problems that justify them.

5.1 Forward Traceability

Problem ID	**Goal ID**	**Objective ID**	**Requirement ID**
P1	G1, G6	IFRO2, IFRO7, INFRO7	FR2, FR9, NFR7
P2	G1, G2	IFRO1, IFRO3, IFRO4	FR3, FR4, FR5, NFR7, FR6
P3	G3	IFRO5, IFRO1, INFRO1	FR7, NFR1, BFR7
P4	G2	IFRO3, IFRO8, INFRO2	FR3, FR10, NFR2
P5	G1, G2	IFRO4	FR4, FR5, FR6
P6	G3, G4	IFRO6, INFRO1	FR8, NFR1
P7	G5	INFRO3, INFRO6	FR11, NFR3, NFR6
P8	G5	INFRO4, INFRO5	NFR4, NFR5
P9	G6	INFRO7	NFR8

5.2 Backward Traceability

5.2.1 Functional Requirements Traceability

Requirement ID	Objective ID	Goal ID	Problem ID
FR1	IFRO1	G1	P2
FR2	IFRO2	G1	P1
FR3	IFRO3	G1, G2	P2, P4
FR4	IFRO4	G1, G2	P2, P5
FR5	IFRO4	G1, G2	P2, P5
FR6	IFRO4	G1, G2	P2, P5
FR7	IFRO5	G3	P3
FR8	IFRO6	G4	P6
FR9	IFRO7	G1	P1
FR10	IFRO8	G2	P4
FR11	INFRO6	G5	P7

5.2.2 Non-Functional Requirements Traceability

Requirement ID	Objective ID	Goal ID	Problem ID
NFR1	INFRO1	G3, G4	P3, P6
NFR2	INFRO2	G2	P4
NFR3	INFRO3	G5	P7
NFR4	INFRO4	G5	P8
NFR5	INFRO5	G5	P8
NFR6	INFRO6	G5	P7
NFR7	INFRO7	G6	P1, P2, P3
NFR8	INFRO7	G6	P9

5.3 Functional and Non-Functional Requirements Relationships

NFR ID	Related Functional Requirements
NFR1	FR2, FR5, FR7
NFR2	FR3
NFR3	FR3
NFR4	FR1, FR5, FR11
NFR5	FR1-FR11
NFR6	FR2, FR7, FR11
NFR7	FR2, FR5, FR6, FR7
NFR8	FR2, FR8, FR10

[6] References

P. Loucopoulos and V. Karakostas, System Requirements Engineering. London, UK: McGraw-Hill, 1995. ISBN: 978-0-07-707843-0.